

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester II Examination, 2023-24

Paper No. BPT-201: Thermal Physics

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1. Answer any **five** of the following:

(a) What do you mean by inversion temperature? How does it depend on van der Waals gas constants? [4]

(b) Prove the following relation for a Carnot cycle:

$$\oint P dV = \oint T dS$$

[4]

(c) Prove that for an isothermal and isobaric process, $dG = 0$, where G is Gibbs' free energy. [4]

(d) Explain the role of temperature in superconductivity. Also explain the behaviour of a superconductor in a magnetic field (Meissner effect). [4]

(e) Explain Kelvin's thermodynamical scale of temperature and define the absolute zero temperature. [4]

(f) Define the efficiency of a Carnot's engine. Find its value if the engine is working between the temperatures of steam (100°C) and ice (0°C). If it absorbs 80 cal of heat, find the amount of heat rejected. [4]

(g) Write down Wien's displacement law and find the wavelength at which human beings radiate maximum energy. (Wien's constant is 0.3 cm-K , human body temperature $\approx 37^\circ\text{C}$). [4]

Question 2.

(a) What do you mean by thermodynamic potentials? By using these potentials, obtain the following Maxwell thermodynamic relations:

$$\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial P}{\partial S}\right)_V \quad \text{and} \quad \left(\frac{\partial T}{\partial P}\right)_S = \left(\frac{\partial V}{\partial S}\right)_P$$

[8]

- (b) Calculate the change in the boiling point of water when the pressure is increased by 1 atmosphere. (Normal boiling temperature of water = 100°C ; Specific volume of steam = $1.671 \text{ m}^3\text{kg}^{-1}$; Latent heat of steam = $2.268 \times 10^6 \text{ J kg}^{-1}$; $1 \text{ atm} = 1.013 \times 10^5 \text{ N/m}^2$). [6]
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Question 3.

- (a) Explain the difference between an ideal and a real gas. Deduce the expression for the Joule-Thomson coefficient and explain the cooling and heating effects for a real gas. [10]
- (b) Calculate the critical constants P_c , V_c , and T_c in terms of van der Waals constants a and b . [4]
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OR

- (a) Describe Kamerlingh Onnes' method of liquefaction of Helium gas with proper details of the apparatus used. Explain the changes in the various properties of liquid helium at temperatures below 2.19 K (λ -transition). [8]
- (b) Describe the principle of regenerative cooling with a schematic diagram of the setup. [6]
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Question 4.

- (a) Write the Maxwell's thermodynamic relations and by using them derive the following relation:

$$C_p - C_v = -T \left(\frac{\partial V}{\partial T} \right)_P^2 \left(\frac{\partial P}{\partial V} \right)_T$$

[8]

- (b) Find the mean free path and the frequency of collision of gas molecules if the coefficient of viscosity of the gas at NTP is 1.66×10^{-4} poises. (Use the gas parameters $v_{rms} = 450 \text{ m/s}$, and density $\rho = 1.25 \text{ g/lit}$). [6]
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OR

- (a) Derive the Clausius-Clapeyron equation for a first-order phase transition and explain the response of Gibbs' function G and its first derivatives w.r.t. temperature and pressure at first-order phase transition. [8]
- (b) Explain the ultraviolet catastrophe in black-body radiation. [6]
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Question 5.

- (a) Derive the following latent heat equation for saturated vapour:

$$C_{sat,2} - C_{sat,1} = \frac{dL}{dT} - \frac{L}{T}$$

where $C_{sat,2}$ and $C_{sat,1}$ are the specific heats of saturated vapour and saturated liquid respectively. [7]

- (b) Derive the following relation for the entropy of a perfect gas:

$$S = C_v \log_e T + (C_p - C_v) \log_e V + \text{constant}$$

[7]